

Customer Shopping Behavior Analysis

1. Project Overview

The **Customer Shopping Behavior Analysis** project focuses on understanding customer purchasing patterns, preferences, and engagement behavior using transactional retail data. The analysis is based on **3,900 customer purchase records** collected across multiple product categories and shopping channels. The primary objective of this project is to transform raw customer data into actionable business insights that can help organizations:

- improve customer engagement,
- optimize marketing strategies,
- enhance customer retention,
- increase sales performance,
- and support data-driven business decisions.

Using **Python, SQL, PostgreSQL, and Power BI**, the project explores customer demographics, spending behavior, subscription trends, product performance, discount usage, and purchasing frequency to identify meaningful trends and business opportunities.

2. Dataset Summary

Dataset Information

Attribute	Details
Total Records	3,900
Total Features	18
Dataset Type	Customer Shopping Transactions
Data Format	CSV

Key Features Included

Customer Demographics

- Age
- Gender
- Location
- Subscription Status

Purchase Information

- Item Purchased
- Product Category
- Purchase Amount
- Season
- Size
- Color

Customer Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

Data Quality Assessment

- Identified **37 missing values** in the review_rating column.
- Performed data cleaning and preprocessing to improve data consistency and analytical accuracy.

3. Exploratory Data Analysis using Python

The initial phase of the project involved data preprocessing, cleaning, transformation, and exploratory analysis using Python.

Technologies & Libraries Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Data Preparation & Cleaning

Data Loading

Imported the dataset into Python using the Pandas library for structured analysis and manipulation.

Initial Data Exploration

Performed:

- df.info() for schema inspection
- df.describe() for statistical summaries
- Null value analysis
- Duplicate record checks

This helped in understanding:

- data types,
- feature distributions,
- and dataset quality.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Missing Value Handling

The review_rating column contained missing values.

Imputation Strategy

Missing review ratings were filled using:

- **Median review rating of each product category**

This approach maintained category-level consistency while minimizing data distortion.

Column Standardization

Renamed all column names into **snake_case** format to improve:

- readability,
- coding consistency,
- and SQL integration.

Example:

- Purchase Amount → purchase_amount

Feature Engineering

Age Group Creation

Generated a new feature called age_group by categorizing customers into demographic age ranges:

- Young Adults

- Adults
- Middle-Aged
- Senior Customers

Purchase Frequency Transformation

Created a new column:

- purchase_frequency_days

This enabled better analysis of customer purchase behavior and loyalty patterns.

Data Consistency Validation

Verified redundancy between:

- discount_applied
- promo_code_used

Since both columns represented similar promotional behavior:

- promo_code_used was removed to reduce redundancy.

Database Integration

Connected the cleaned dataset to:

PostgreSQL

using:

pgAdmin 4

The transformed dataset was loaded into PostgreSQL for advanced SQL-based business analysis.

4. Data Analysis using SQL (Business Transactions)

Structured SQL analysis was performed to answer key business questions and extract valuable customer insights.

1. Revenue Analysis by Gender

Compared total revenue contribution between:

- Male customers
- Female customers

Objective

Identify high-value customer demographics and purchasing behavior trends.

	gender 	revenue 
	text	numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users

Identified customers who:

- used discounts,
- but still spent above the average purchase amount.

Business Insight

These customers represent:

- strong purchasing intent,
- and high-value promotional targets.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:01

3. Top 5 Products by Review Rating

Analyzed products with the highest average customer ratings.

Objective

Identify:

- highly preferred products,
- and customer satisfaction drivers.

	item_purchased 	Average Product Rating 
	text	numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts across:

- Standard Shipping
- Express Shipping

Insight

Express shipping users showed higher purchasing tendencies, indicating urgency-driven buying behavior.

	shipping_type 	round 
	text	numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs Non-Subscribers Analysis

Compared:

- average spending,
- total revenue,
- and customer engagement

between:

- subscribed customers,
- and non-subscribers.

Key Finding

Subscribers contributed significantly higher long-term value and repeat purchases.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified products with the highest percentage of discounted purchases.

Purpose

Understand:

- price-sensitive products,
- and discount-driven buying patterns.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Segmented customers into:

- New Customers
- Returning Customers
- Loyal Customers

based on purchase history and transaction frequency.

Business Value

Helps create:

- personalized marketing campaigns,
- loyalty programs,
- and retention strategies.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top Products per Category

Identified the top 3 most purchased products within each product category.

Objective

Support:

- inventory planning,
- product promotion,
- and merchandising strategies.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscription Behavior

Analyzed whether customers with more than 5 purchases were more likely to subscribe.

Insight

Frequent buyers showed stronger subscription adoption patterns.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue Contribution by Age Group

Calculated revenue generated by different customer age groups.

Purpose

Identify:

- high-revenue demographics,
- and targeted marketing opportunities.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Interactive Dashboard using Power BI

An interactive business intelligence dashboard was developed using:

Microsoft Power BI

The dashboard provides stakeholders with dynamic visual insights into customer behavior and sales performance.



Dashboard Features

KPI Metrics

- Total Revenue
- Average Purchase Amount
- Subscription Count
- Repeat Customers

Interactive Visualizations

- Revenue by Gender
- Age Group Analysis
- Product Category Performance
- Seasonal Sales Trends
- Discount Impact Analysis
- Shipping Preference Trends

User Interaction

- Dynamic filters
- Drill-down analysis
- Category-wise segmentation
- Customer behavior tracking

6. Key Business Insights

Customer Loyalty Drives Revenue

Returning and loyal customers generated significantly higher revenue compared to new customers.

Subscribers Are More Valuable

Subscribed customers demonstrated:

- higher average spending,
- stronger engagement,
- and increased repeat purchases.

Discounts Influence Purchasing Decisions

Several products showed strong dependency on discounts, indicating price-sensitive buying behavior.

Express Shipping Users Spend More

Customers selecting express shipping generally placed higher-value orders.

Certain Age Groups Dominate Revenue

Specific age segments contributed the majority of revenue, making them ideal targets for marketing campaigns.

7. Business Recommendations

1. Strengthen Subscription Programs

Introduce:

- exclusive offers,
- early access discounts,
- and loyalty rewards

to increase subscription adoption and customer retention.

2. Develop Customer Loyalty Programs

Reward repeat customers through:

- cashback,
- reward points,
- referral incentives,
- and personalized offers.

This can help transition customers into the “Loyal” segment.

3. Optimize Discount Strategies

Implement data-driven discount policies to:

- maximize conversions,
- while protecting profit margins.

4. Promote High-Performing Products

Use top-rated and best-selling products in:

- advertisements,
- recommendations,

- and promotional campaigns.

5. Enable Targeted Marketing

Focus marketing efforts on:

- high-revenue age groups,
- loyal customers,
- and premium shipping users.

8. Conclusion

This project demonstrates the practical application of data analytics in understanding consumer shopping behavior and improving business performance.

By combining:

- Python for data preprocessing,
- SQL for structured business analysis,
- and Power BI for interactive visualization,

the project successfully transformed raw customer transaction data into meaningful business insights.

The findings can help organizations:

- improve customer engagement,
- optimize sales strategies,
- enhance customer loyalty,
- and make data-driven business decisions effectively.